Cayetano Hernández Sánchez

Physicist | Scientific Computing & Data Analysis

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Profile

Physicist (MSc student, Uppsala University) specialised in scientific computing in Python, Fortran and C++, from simulating electron transport in disordered materials to Bayesian analysis of stellar spectra on a supercomputer.

Education

2025 – present	Uppsala University: Master's in Physics, Astronomy and Space Physics
2025	University of Valencia: Expert in Observational Astrophysics
2019 – 2024	University of Murcia: Bachelor in Physics

Technical Skills

Programming	**Advanced:** Python (NumPy, SciPy, pandas, Matplotlib), Fortran, C++ **Intermediate:** SQL, R, MATLAB, IDL
Methods	Numerical simulation, HPC, Bayesian inference (MCMC), statistical model fitting
Tools	Git, LaTeX, Excel, Word; image processing with AstroImageJ and PixInsight
Physics	Condensed matter, statistical and quantum physics, electromagnetism, astrophysics

Computational Projects

2026 – present **Detecting Lithium-6 in a very metal-poor star** (Master project, now thesis)

- Ran and benchmarked 64 3D radiative-transfer simulations on the Dardel supercomputer, cutting compute cost while keeping numerical errors 100× below the signal.
- Built a Python pipeline (normalisation, interpolation, MCMC with `emcee`) to fit a weak signal in noisy spectra with strongly degenerate parameters.
- Reanalysed a published result with 3D models; aimed at a first-author paper.

2024 **Variable Range Hopping simulations in granular metals** (Bachelor thesis)

- Metropolis–Hastings Monte Carlo code in Fortran 90 for hopping in granular metals.
- Conductivity vs. temperature for three disorder ranges: activated law vs. softer law.

Employment History

2024 **Research Assistant**, Uppsala University

- Simulated pre-supernova neutrino detection rates in JUNO, KamLAND and SK-Gd.

2024 **Private Tutor**: Physics, chemistry and maths (pre-university level)

- Step-by-step problem solving and clear explanations adapted to each student.

2021 **Data Analyst Intern**, Artificial Intelligence Talentum

- Data cleaning, exploratory analysis and visualisation.

Languages & Others

Languages	**Spanish** (native), **English** (Cambridge C1 Advanced), **German** (A1, Goethe)
2020, 2021, 2023	**Science Outreach**: GTC *PaCienciaLaNuestra*, Murcia Science Week
2019	**Extraordinary Baccalaureate Prize Nominee**, Valencian Community

References

Dr. Andreas Korn, Uppsala University
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Dr. Karin Lind, Stockholm University
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